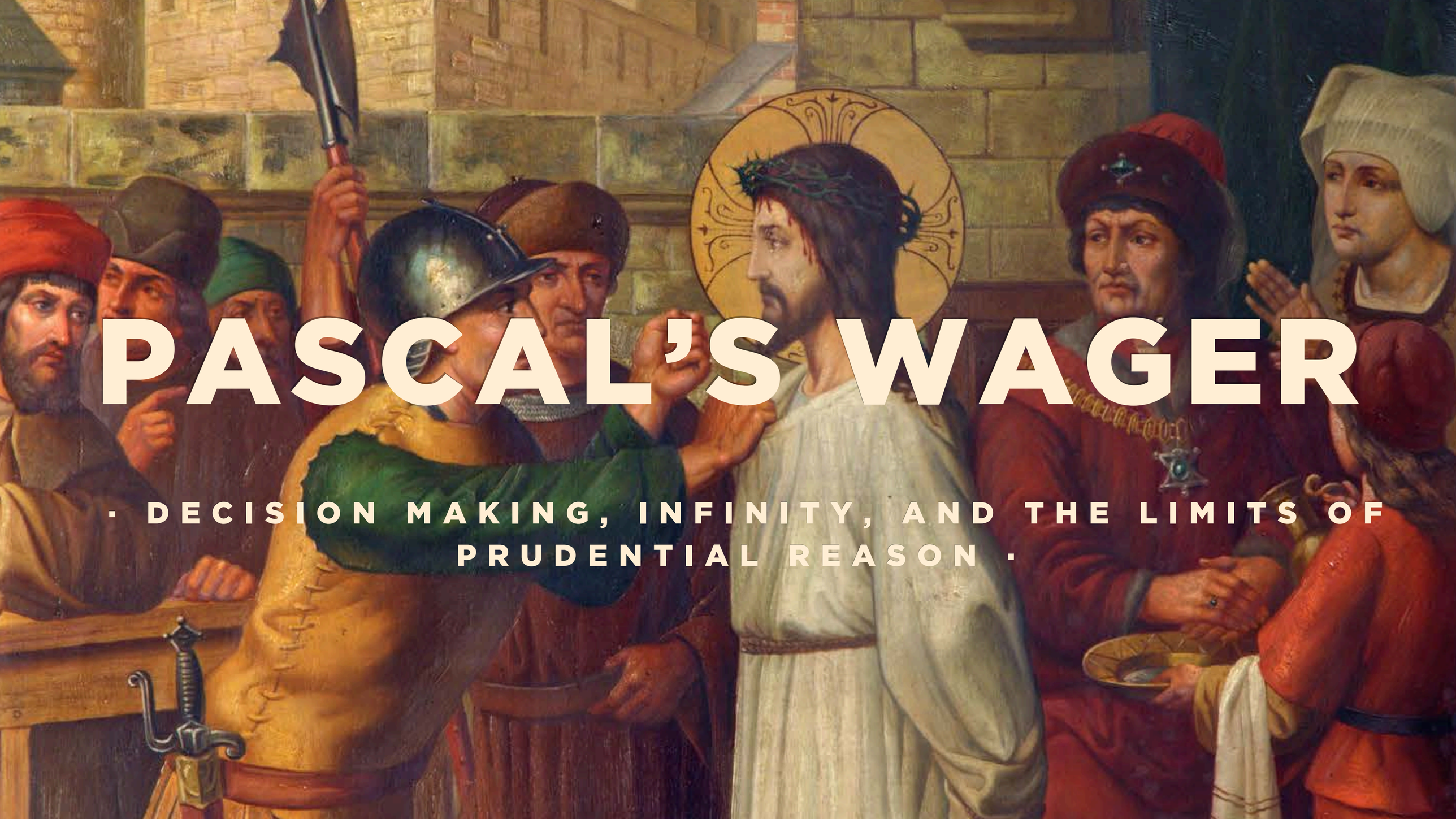
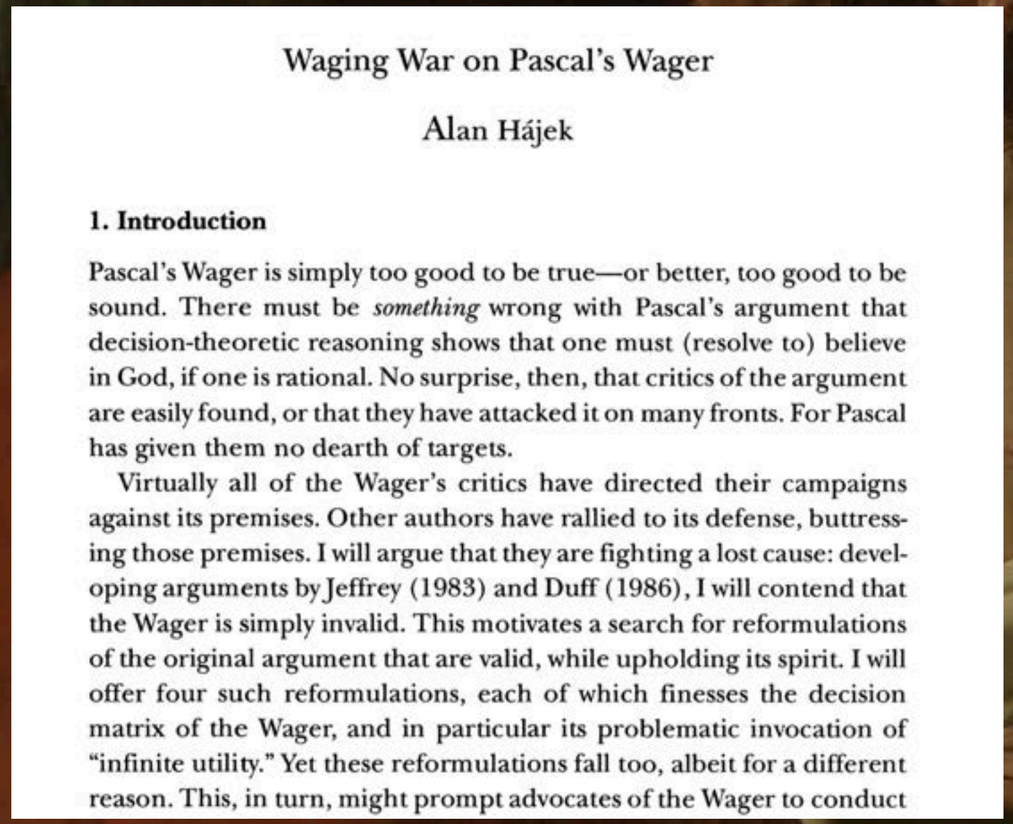


PASCAL'S WAGER

· DECISION MAKING, INFINITY, AND THE LIMITS OF PRUDENTIAL REASON ·

SOURCES

Waging War on Pascal's Wager
Alan Hájek
The Philosophical Review, Vol. 112, No. 1 (Jan., 2003), pp. 27-56 (30 pages)
https://www.jstor.org/stable/3595561



Waging War on Pascal's Wager

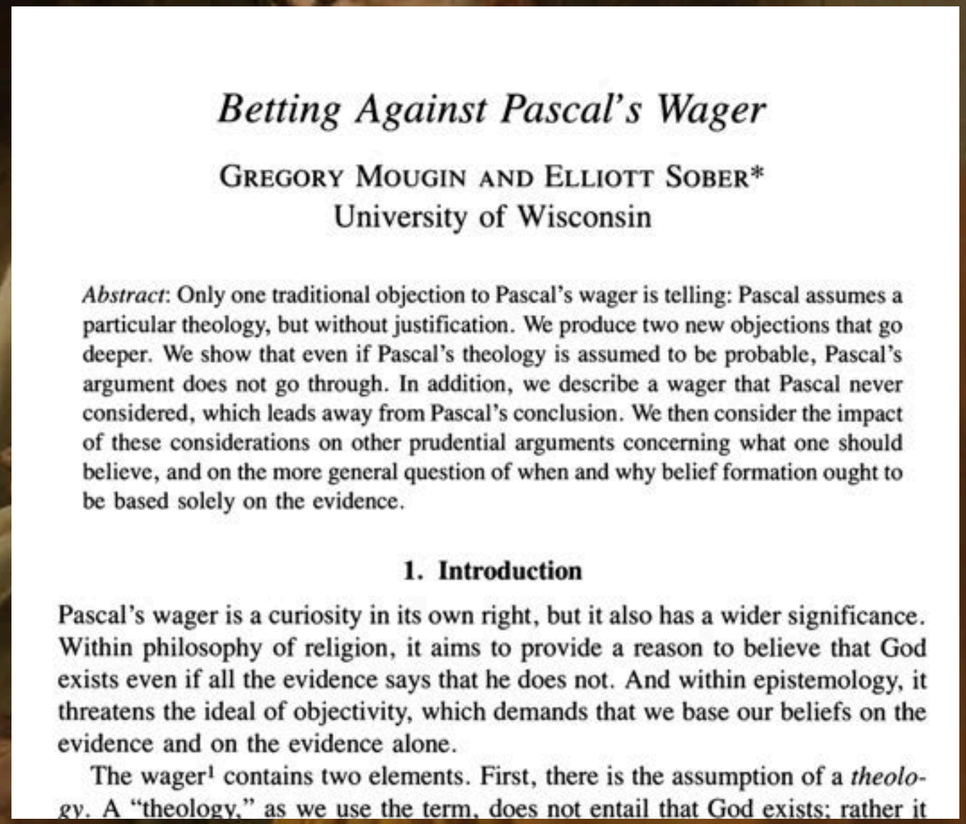
Alan Hájek

1. Introduction

Pascal's Wager is simply too good to be true—or better, too good to be sound. There must be *something* wrong with Pascal's argument that decision-theoretic reasoning shows that one must (resolve to) believe in God, if one is rational. No surprise, then, that critics of the argument are easily found, or that they have attacked it on many fronts. For Pascal has given them no dearth of targets.

Virtually all of the Wager's critics have directed their campaigns against its premises. Other authors have rallied to its defense, buttressing those premises. I will argue that they are fighting a lost cause: developing arguments by Jeffrey (1983) and Duff (1986), I will contend that the Wager is simply invalid. This motivates a search for reformulations of the original argument that are valid, while upholding its spirit. I will offer four such reformulations, each of which finesses the decision matrix of the Wager, and in particular its problematic invocation of "infinite utility." Yet these reformulations fall too, albeit for a different reason. This, in turn, might prompt advocates of the Wager to conduct

Betting Against Pascal's Wager
Gregory Mougin, Elliott Sober
Noûs, Vol. 28, No. 3 (Sep., 1994), pp. 382-395 (14 pages)
https://doi.org/10.2307/2216065•https://www.jstor.org/stable/2216065



Betting Against Pascal's Wager

GREGORY MOUGIN AND ELLIOTT SOBER*
University of Wisconsin

Abstract: Only one traditional objection to Pascal's wager is telling: Pascal assumes a particular theology, but without justification. We produce two new objections that go deeper. We show that even if Pascal's theology is assumed to be probable, Pascal's argument does not go through. In addition, we describe a wager that Pascal never considered, which leads away from Pascal's conclusion. We then consider the impact of these considerations on other prudential arguments concerning what one should believe, and on the more general question of when and why belief formation ought to be based solely on the evidence.

1. Introduction

Pascal's wager is a curiosity in its own right, but it also has a wider significance. Within philosophy of religion, it aims to provide a reason to believe that God exists even if all the evidence says that he does not. And within epistemology, it threatens the ideal of objectivity, which demands that we base our beliefs on the evidence and on the evidence alone.

The wager[1] contains two elements. First, there is the assumption of a *theology*. A "theology," as we use the term, does not entail that God exists; rather it

PASCAL'S WAGER

PASCAL'S WAGER

343 [6–233]

Infinity. Nothingness. Our soul has been cast into the body, where it finds number, time and dimension. It reasons thereupon, and calls it nature, necessity, and can believe nothing else.

Unity added to infinity adds nothing to it, any more than does one foot added to infinite length. The finite is annihilated in presence of the infinite, and becomes pure nothingness. So does our mind before God; so does our justice before divine justice.

There is not so great a disproportion between human and divine justice as between unity and infinity.

The justice of God must be as vast as His mercy. But his justice done upon the reprobate is not so vast as, and should shock us less than, His mercy shown towards the elect.

We know that the infinite exists, but we are ignorant of its nature. Since we know it is false to say that number is finite, it must be true that there is infinity in number. But we do not know what it is. We cannot say that it is even, or that it is odd. Yet it is a number, and every number is either even or odd (this is certainly true of every finite number). So we may perfectly well know that God exists, without knowing what He is.

Is there not one substantial truth, seeing that there are so many things which are not truth itself?

We know then the existence and nature of the finite, because we too are finite and have extension. We know the existence of the infinite, but not its nature; for, like us, it has extension but no limits such as we have. But we know neither the existence nor the nature of God, because He has neither extension or limits.

But by faith we know His existence; in the light of glory we shall know His nature. I have already shown that there is nothing to prevent our knowing the existence of a thing, without knowing its nature.

PASCALS WAGER

To Pascal, belief in god is a **decision problem** under **uncertain conditions**

Since (to Pascal) reason alone fails to settle the matter of god's existance, Pascal instead **appeals to self interest.**

Faith is therefore transformed into a wager between a finite stake and infinite potential gain.

Two elements constitute the wager:

- a **theology** – conditional claims about god's actions should he exist
- a **decision matrix** – payoffs assigned to the binary choice between theism and atheism against the two possible states of the world

"Either God exists, or He does not. To which view shall we incline? Reason cannot decide for us one way or the other (Pascal, Pensées (1662))."

Pascal, Pensées (1662) · Mougin & Sober, Noûs (1994) · Hájek, Philosophical Review (2003)

Pascal, Pensées §233; Mougin & Sober (1994), p. 382

THE DECISION MATRIX

	Exists	¬ Exists
Theism	heaven - infinite gain	modest loss
¬ Theism	hell - infinite loss	modest gain

if [(theism ∧ exists) - (¬ theism∧exists)]:= P(i) is infinite and [(¬theism∧¬exists) - (theism∧¬exists)] := P(j) if finite, theism dominates if Pr(god) > 0

U(theism) > U(atheism) iff Pr(P(i)) > (1-Pr(P(j)))

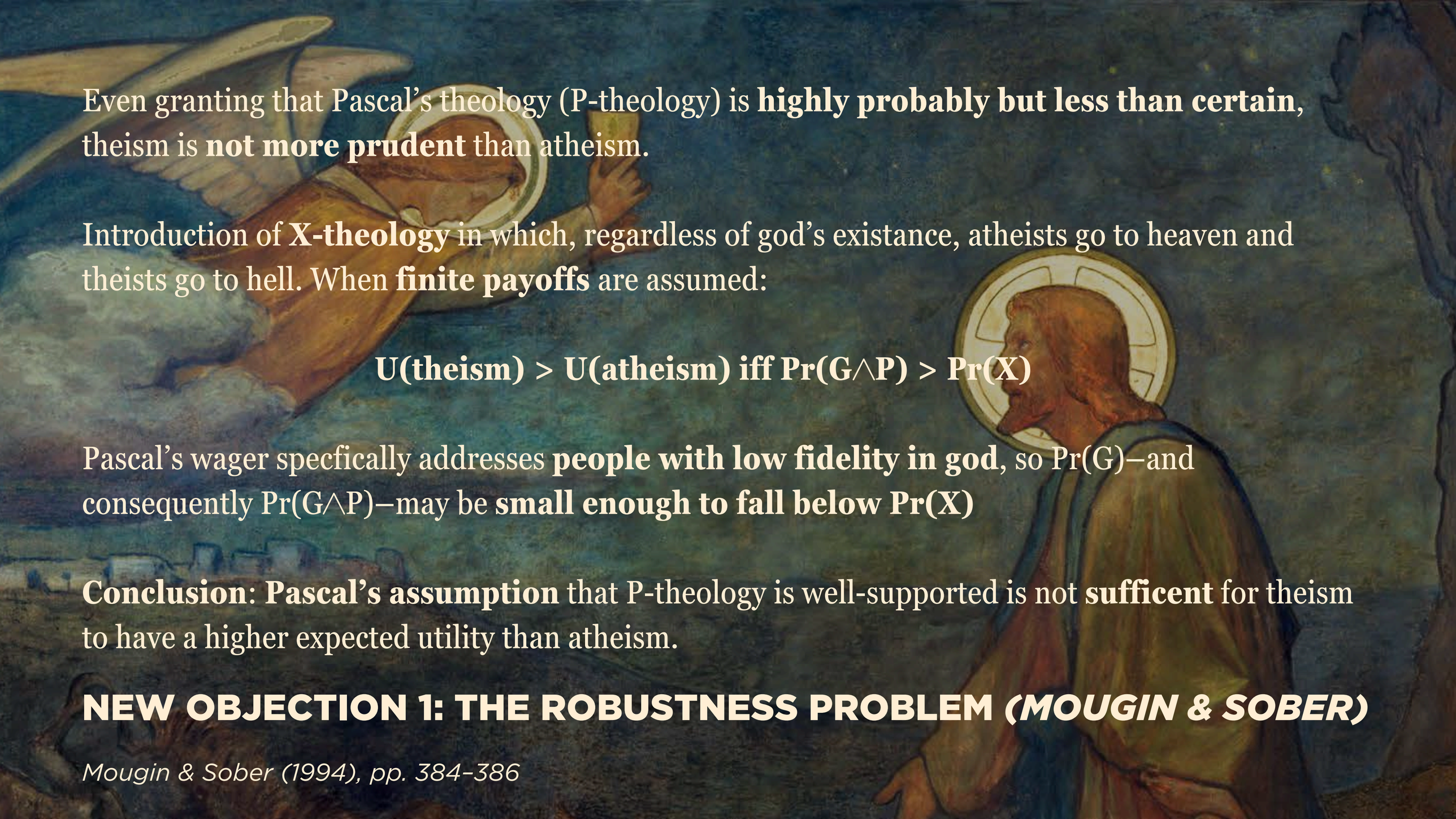
THREE TRADITIONAL OBJECTIONS

A **The Evidentialist Objection (Clifford):** It is “wrong always, everywhere, and for anyone, to believe anything upon insufficient evidence.” Mougin & Sober reply Clifford’s argument for objectivity is prudential.

B **Belief Is Not a Voluntary Act:** One cannot simply decide to believe. Pascal anticipates this: he advises acting as if one believes — attending mass, living among believers — so that faith gradually takes hold. Mougin & Sober accept Pascal’s reply but question just how contagious belief is for confirmed skeptics.

C **The Unjustified Theology Objection:** Pascal provides no justification for his specific theological assumptions. Mougin & Sober find this the most compelling of the three.

Mougin & Sober (1994), pp. 383–384



Even granting that Pascal's theology (P-theology) is **highly probably but less than certain**, theism is **not more prudent** than atheism.

Introduction of **X-theology** in which, regardless of god's existance, atheists go to heaven and theists go to hell. When **finite payoffs** are assumed:

$$\mathbf{U(theism) > U(atheism) \text{ iff } Pr(G \wedge P) > Pr(X)}$$

Pascal's wager specfically addresses **people with low fidelity in god**, so Pr(G)—and consequently Pr(G∧P)—may be **small enough to fall below Pr(X)**

Conclusion: **Pascal's assumption** that P-theology is well-supported is not **sufficent** for theism to have a higher expected utility than atheism.

NEW OBJECTION 1: THE ROBUSTNESS PROBLEM *(MOUGIN & SOBER)*

Mougin & Sober (1994), pp. 384-386

NEW OBJECTION 2: PASCAL'S LOCAL PRUDENTIALISM (MOUGIN & SOBER)

If belief in God can be revised on prudential grounds, why not theology too?

For an atheist who accepts P-theology evidentially, X-theology offers a prudential advantage: believing X-theology removes the anxiety of thinking one might be punished if God exists and one is wrong.

- ***Atheist believing P-theology**: rots + psychological burden of potential divine punishment*
- ***Atheist believing X-theology**: rots, no such anxiety*

*The atheist's natural starting point is **prudentially unstable**. Pascal's "local prudentialism," which subjects only belief in God to prudential scrutiny, is **arbitrarily narrow***

Mougin & Sober (1994), pp. 387-388

THE WAGER IS SIMPLY INVALID *(HÁJEK)*

Even granting Pascal all his premises, wagering for God is not uniquely rationally mandated.

The mixed-strategy problem: toss a fair coin and wager for God only if it lands heads. By Pascal's reasoning, this strategy's expected utility is:

$$\tfrac{1}{2}(\infty) + \tfrac{1}{2}(\text{finite}) = \infty$$

This is equal to outright wagering for God. The same holds for infinitely many mixed strategies. Pascal did not consider them, but decision theory he employs **includes them automatically.**

Consequence: by Pascal's theology, **every rational agent's every action has maximal expected utility.** The Wager collapses into a **proof that all practical reasoning is equally good** - a reductio of its own framework

Hájek (2003), pp. 30–33

REFORMULATIONS AND THE CORE DILEMMA (HÁJEK)

Horn 1: If the utility of salvation is reflexive under both addition and multiplication by positive finite probabilities (def. of ∞), the Wager is invalid: all mixed strategies tie in expectation.

Hájek offers four reformulations — surreal utility, vector-valued value, long-run average utility, and finite utility — each solving the mixed-strategy problem, yet all face the same dilemmas

"I wager that any future version of the argument will succumb to this dilemma Hájek (2003)."

Horn 2: If the utility of salvation is not reflexive under addition or multiplication, salvation is far from being the best possible thing — it is swamped infinitely many times over — and is therefore unsuitable for Pascal's theology.

Hájek (2003), pp. 45–49

BROADER SIGNIFICANCE: OBJECTIVITY AND PRUDENTIAL BELIEF

Does Pascal's failure mean prudential belief-revision is always wrong?

Mougin & Sober say no: prudential considerations can **robustly** override evidence — when a belief's revision survives scrutiny of the background assumptions. Pascal's wager fails this test: subjecting P-theology to the same prudential logic undermines the conclusion.

Hájek's dilemma: the problem is not merely due to incorrect theology or probability. The structure of infinite utility in decision theory is itself untenable, and no reformulation escapes.

The open question: under what general conditions is it prudent to be objective — to base belief on evidence alone? Pascal's Wager marks an outer boundary where prudential reasoning fails.

Mougin & Sober (1994), pp. 389–393; Hájek (2003), pp. 48–49